

EEE313 ENGINEERING PROJECT REPORT

VGA-Interfaced 3-Bit ADC on Basys3

Discrete-MOSFET analog front end, Three-comparison converter, Trigger-aligned capture system, and VGA waveform display

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Repository: <https://github.com/Borek-32/EEE313-VGA-Scope-3-Bit-ADC-BASYS3>

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Abstract

This project implements a complete three-bit analog-to-digital converter on a breadboard and displays the sampled waveform on a VGA monitor. The assignment called for a breadboard implementation with Arduino; I used a Digilent Basys3 FPGA to control conversion and generate the display directly. The analog circuit uses a BS250 sampling switch, a 33 nF hold capacitor, a five-transistor comparison stage, CMOS regeneration, and a resistor ladder selected by a multiplexer. The FPGA performs three successive comparisons, detects a repeatable waveform period, and captures the resulting codes in alternating display buffers. A 48 μ s conversion period gives a nominal sampling rate of 20.833 kSa/s. LTspice simulations explain the sample-and-hold operation, comparator polarity, and reference levels. In the laboratory presentation, a signal generator supplied sine, square, triangle, and sawtooth inputs while the oscilloscope and VGA monitor showed the input and converted output together. The tests produced stable, recognizable three-bit traces at 1 kHz and demonstrated capture at 500 Hz and 200 Hz. The completed system met the project objective of building and demonstrating a breadboard ADC, with Basys3 providing conversion control, triggering, storage, and a 640 $\times$ 480 VGA display.

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1 Introduction and Project Objective

The project objective was to construct a three-bit ADC on a breadboard and demonstrate conversion of an analog waveform in the laboratory. The assignment proposed Arduino for digital control. I implemented that role with Basys3 and added a VGA display, so the sampled waveform could be viewed directly on a monitor. The completed setup combines the breadboard analog circuit, FPGA, signal generator, and oscilloscope. At the nominal test condition, it converts a 1 kHz, 2 V_{pp} input into an eight-level waveform and holds the displayed trace at a repeatable phase.

The physical circumstances determined the block choices. A sample-and-hold was necessary because the input continues moving while a sequential converter decides three bits. A compact MOSFET differential stage supplies the required polarity decision, and a following inverter converts its gradual output into a decisive logic level. The reference ladder is concentrated inside the expected signal window because spreading only eight codes across the full 0–5 V supply would waste resolution. On the digital side, threshold hysteresis fixes the capture phase and alternating buffers prevent the visible trace from being overwritten during a scan.

1.1 Project overview

Table 1. Project objectives, implemented responses, and demonstrated results.

Project objective	Implemented response	Project result
Freeze a moving input during three comparisons	BS250 switch, 33 nF hold capacitor, 48 μ s conversion cadence	Schematic, LTspice transient, and held waveform on the oscilloscope
Make a one-bit threshold decision	5T OTA-like stage followed by CMOS regeneration	Topology, AC screening, inverter transient, threshold sweep
Use the three-bit range where the signal moves	Eight references from 2.556 V to 4.500 V	Ladder source, static LTspice result, voltage table, meter spot-check
Create one code from three comparator decisions	Four-state controller with three settled comparison stages	VHDL controller and sampling state diagram
Keep the waveform stationary	Low/high hysteresis, full-period detection, short holdoff	Trigger VHDL, planning diagram, trigger/capture flow
Display without overwriting the visible trace	Two 256 $\times$ 3-bit buffers and a 25 MHz VGA renderer	Elaborated design, VHDL, and programmed-board waveform demonstrations

The laboratory presentation exercises the complete path from the signal generator through the breadboard converter to the VGA monitor. Sine, square, triangle, and sawtooth tests show how the converter responds to different input shapes. The circuit schematics, simulation plots, and laboratory photographs explain the design and its measured behavior throughout this report.

2 Design Context and Approach

Several assignment circumstances shaped the architecture. The input was centered around 3.5 V rather than ground, the analog building blocks were expected to use the available BS250 and 2N7000 MOSFETs, the external circuitry operated around 5 V while Basys3 I/O uses 3.3 V LVCMOS, and the conversion had to finish quickly enough to make a 1 kHz waveform recognizable. These constraints explain why the design contains custom sampling, regeneration, and translation stages instead of an integrated converter or comparator.

The conversion period is 48 μs . The LTspice test stimulus allocates 12 μs to tracking and 36 μs to holding; the VHDL controller allocates approximately 8 μs to tracking and 40 μs to holding. In each case, the capacitor acquires the input before the comparison interval. The phase allocation determines the settling time available to the switch and the duration over which the capacitor holds the sample.

One 48 μs cadence, two recorded phase allocations

Green = track/acquire; gray = hold/compare

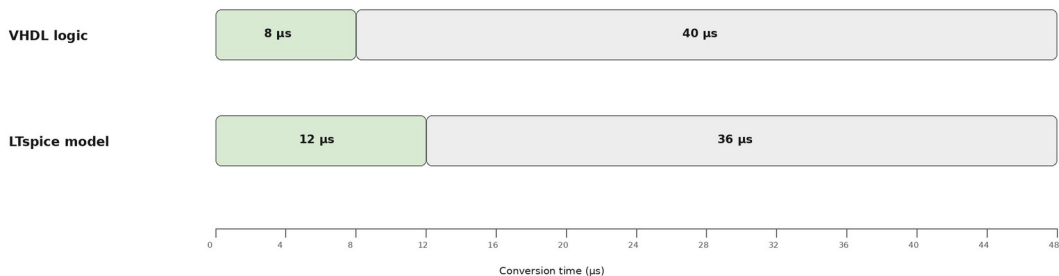


Figure 1. One conversion cadence and the two recorded phase allocations.

Both timing descriptions use a 48 μs frame, corresponding to 20.833 kSa/s. The analog simulation quantifies settling and droop for its 12/36 μs stimulus, while the controller sequence determines the sampling cadence of the FPGA implementation.

Table 2. Design circumstances and the decisions they produced.

Circumstance	Decision I made	Reason
Sequential three-bit decision	Sample and hold ahead of the comparator	All three bit decisions must refer to one input sample
Discrete MOSFET constraint	5T OTA-like stage plus inverter	Separate threshold sensitivity from rail regeneration using available devices

Circumstance	Decision I made	Reason
3.5 V-centered input	Compressed reference ladder	Spend the eight codes across the useful voltage interval
3.3 V FPGA / 5 V analog boundary	Two cascaded CMOS inverters	Translate the control voltage while preserving logical polarity
Unstable free-running display	Hysteretic trigger and two-period capture	Begin each frame at a repeatable phase and show a complete segment
Capture continues while VGA scans	Inactive-buffer write and completed-buffer swap	Avoid writing the memory currently being rendered

I simulated the analog stages in LTspice, assembled them on the breadboard, and tested the complete system with the laboratory signal generator. Oscilloscope captures show the held waveform and control activity. The presentation photographs place the generator, oscilloscope, and VGA output together, making the applied input and converted result directly comparable. The VHDL and Vivado hierarchy explain how the FPGA controls conversion and draws the output.

Table 3. How each part of the design was checked.

Material used	What it shows	How it is used
LTspice circuit and plots	Analog topology, sample/hold behavior, reference spacing, and comparator polarity	Sizes the analog stages and quantifies the modeled response
VHDL and pin constraints	Controller, capture, VGA, polarity, and pin behavior	Defines the implemented digital signal path
Vivado hierarchy	Clock generator, ADC controller, and VGA module	Shows how the digital modules connect
Oscilloscope and meter photographs	Held-waveform behavior, control activity, and a reference-voltage reading	Connects the analog calculations with bench operation
VGA photographs	Recognizable sine, square, triangle, sawtooth, and frequency-limit behavior	Shows the complete acquisition and display path
Planning diagrams	Binary-search, trigger, and architecture concepts	Introduces the control decisions expanded by the VHDL chapters

3 Hardware Platform and User Operation

The Digilent Basys 3 was selected because its XC7A35T FPGA, 100 MHz oscillator, PMOD headers, LEDs, and VGA connector let one board coordinate the external converter and display the result directly. The analog circuitry operates around 5 V, while the constrained FPGA pins are LVCMOS33. The outgoing-control and comparator-return paths are therefore treated as separate electrical boundaries.

top_adc produces logical HOLD and natural-binary reference-selection values. The control interface transfers these commands through the PMOD connection to the breadboard sample switch and reference multiplexer. The comparator returns one decision bit on din. During the demonstration, the signal generator drives the analog input, the oscilloscope displays the test signal, and the VGA monitor displays the converted samples. The three LEDs also show the latest ADC code.

Table 4. Hardware controls and observations used during normal operation.

User-facing element	Implemented role	Operating reason
Analog input	Nominal 1 kHz, 2 Vpp waveform centered near 3.5 V	Keeps the signal inside the compressed reference window
PMOD JB	HOLD and SEL[2:0] control outputs	Connects conversion control to the breadboard level converters
din return	One-bit comparator decision on constrained pin K17	Closes the analog-to-digital decision loop
LED[2:0]	Latest completed ADC code	Provides immediate board-level state visibility
VGA connector	640 × 480 trigger-stable waveform	Makes the complete acquisition result visible without a host application

The comparator output returns to din on K17. This one-bit signal closes the conversion loop: each comparison selects the next reference and contributes one bit to the completed ADC code.

4 System Architecture and Data Contracts

The system boundary is the one-bit comparison event. The analog side receives the waveform, holds one value, selects a reference, and returns din. Inside the FPGA, top_adc generates active-high logical controls and assembles three comparator decisions. top_main connects HOLD and SEL to the PMOD outputs, and the two-stage level converters preserve their polarity while translating them to the 5 V analog domain. Completed codes then pass through period detection, alternating capture buffers, and VGA rendering.

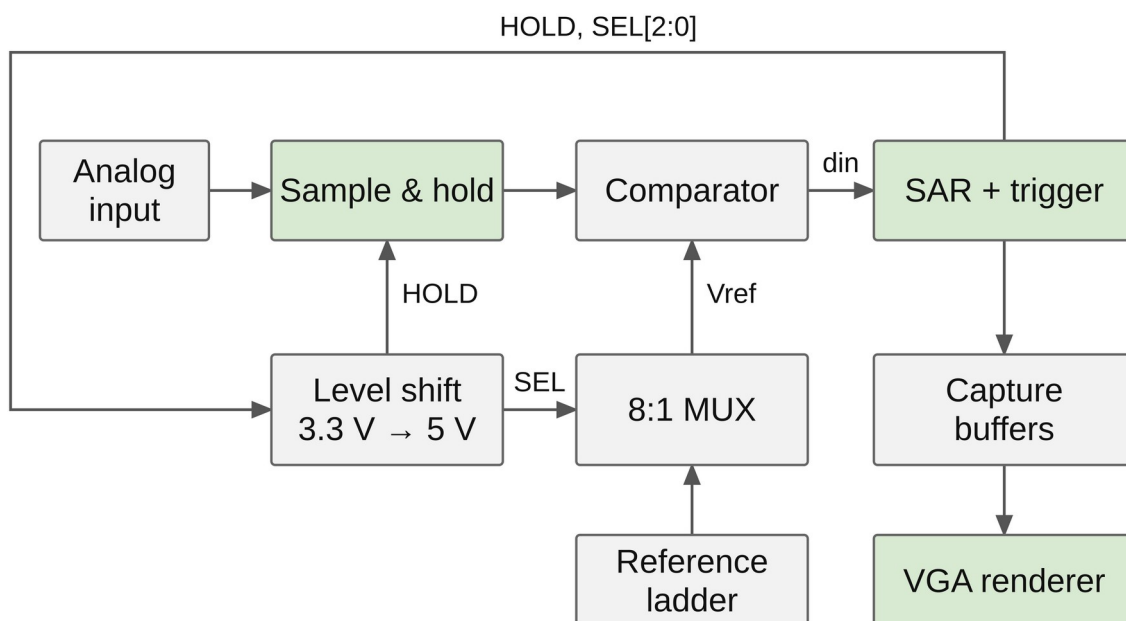


Figure 2. Implemented mixed-signal signal path and control loop.

The control loop crosses the voltage interface in both directions. HOLD and SEL travel from the FPGA to the 5 V analog circuit through the level converters. The comparator decision returns through the 5 V-to-3.3 V interface to the FPGA input.

4.1 Interfaces and data contracts

The system has three useful rates. The board clock is 100 MHz. The Clocking Wizard produces 25 MHz for VGA. A divider in `top_adc` toggles `dclk` at 250 kHz, and the four-state controller consumes twelve rising `dclk` edges per completed code. The `code_valid` and `period_valid` events originate with the conversion logic and are synchronized into the 100 MHz capture process. The visible trace is rendered from a completed buffer in the 25 MHz domain.

Table 5. Principal interfaces and the contract carried by each signal.

Interface	Direction	Meaning at the boundary
hold / sel[2:0]	top_adc → top_main	Active-high HOLD and natural-binary selector values
PMOD control outputs	FPGA → level converters	3.3 V commands for the sample switch and reference multiplexer
HOLD / SEL at 5 V	level converters → analog	Control polarity preserved by two cascaded inverters
din	analog → FPGA	One-bit result: held input above or below the selected reference
code_out[2:0]	ADC controller → capture	Completed three-bit sample
code_valid	ADC controller → capture	One event per completed code

Interface	Direction	Meaning at the boundary
period_valid	trigger → capture	Repeatable phase reference after a complete low-to-high cycle
disp0 / disp1	capture ↔ renderer	One buffer is written while the other remains visible

5 Analog Acquisition and Comparison Path

The analog design begins with the timing problem created by a sequential converter. For a 1 kHz sine wave with 1 V peak amplitude, the maximum slope is approximately $2\pi fA = 6.28 \text{ kV/s}$. Across one $48 \mu\text{s}$ conversion frame, the input can move by roughly 0.30 V, which is comparable to the 0.278 V reference spacing. Three direct comparisons could therefore describe three different input values; the sample-and-hold removes that ambiguity.

$$\text{maximum input change over } 48 \mu\text{s} \approx 2\pi(1 \text{ kHz})(1 \text{ V})(48 \mu\text{s}) \approx 0.302 \text{ V}$$

5.1 Sample-and-hold design

Because the analog section had to use discrete MOSFET building blocks, I used a BS250 as the sampling switch and stored the input on 33 nF. The first-order sizing used a 50Ω source and approximately 10Ω switch resistance, giving a nominal $1.98 \mu\text{s}$ acquisition time constant. A three-bit half-LSB settling target corresponds to 1/16 of full scale. The 33 nF value therefore gives useful settling margin during the available acquisition interval while keeping simulated hold drift small during the comparison window.

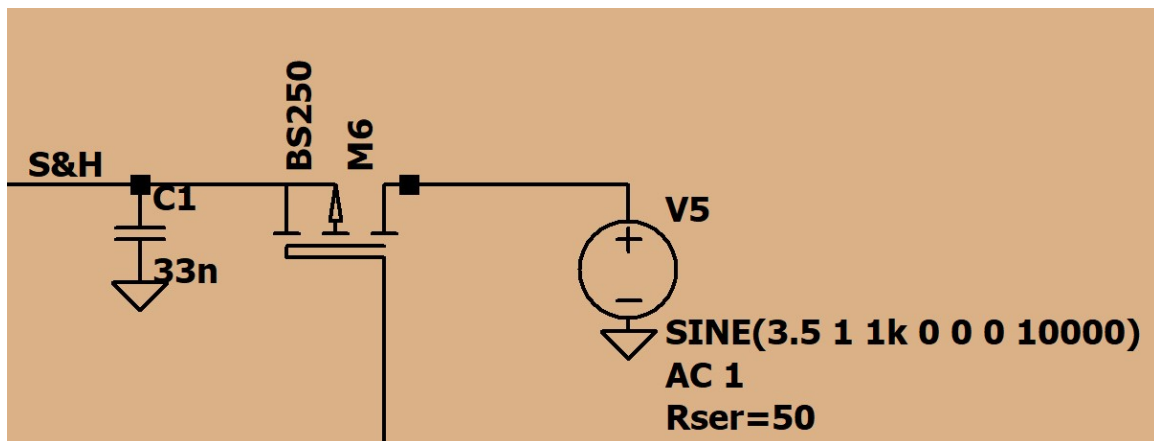


Figure 3. MOSFET sample-and-hold used to freeze the input during conversion.

The BS250 switch and 33 nF capacitor provide the analog memory needed by the sequential conversion. This low-component-count topology fits the available device constraint and the three-bit resolution target. The transient analysis below quantifies its settling and droop.

The transient contains 208 complete acquisition and hold intervals. The nominal input spans 2.4997–4.5000 V and the held node spans 2.5063–4.4995 V. The 12.75 mV maximum acquisition endpoint error is about 4.59% of one 0.278 V ladder interval, while the 0.461 mV maximum

endpoint drift is about 0.166% of one interval. These values support the 33 nF choice for the modeled condition.

LTspice nominal transient analysis

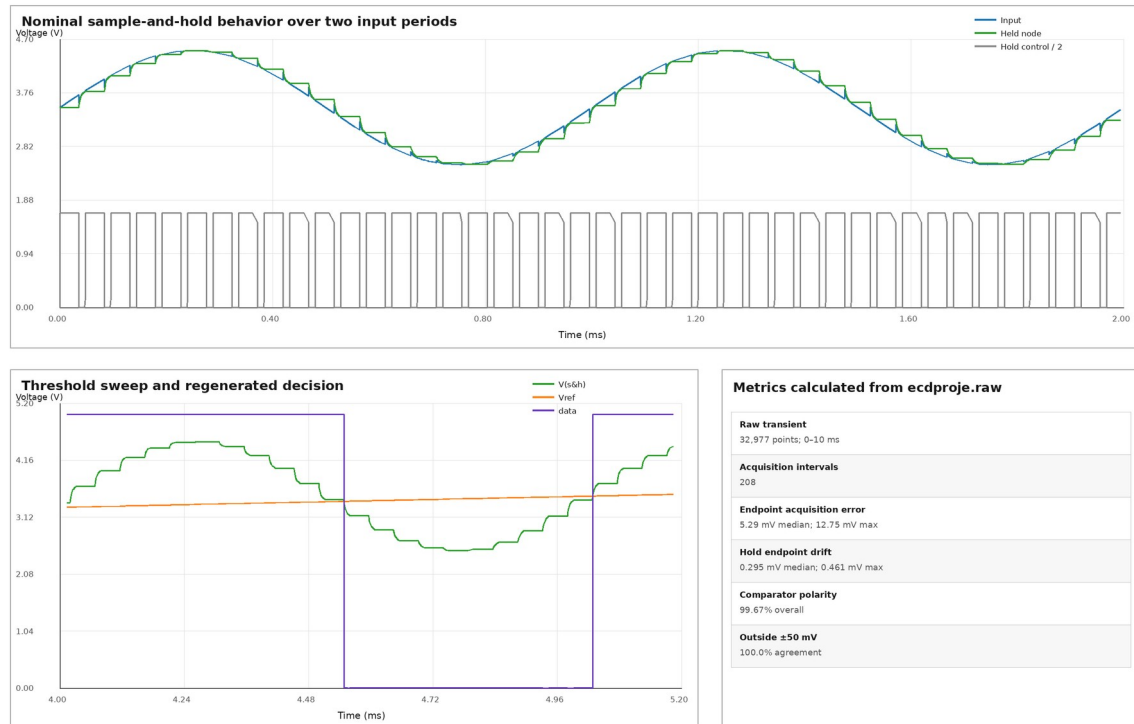


Figure 4. LTspice transient, threshold behavior, and nominal metrics.

Table 6. Nominal LTspice metrics computed from the RAW transient.

Simulation quantity	Nominal result	Meaning
Input range	2.4997–4.5000 V	Covers the intended $3.5\text{ V} \pm 1\text{ V}$ stimulus
Held-node range	2.5063–4.4995 V	Tracks nearly the complete signal window
Acquisition endpoint error	5.29 mV median; 12.75 mV max	Settling at the end of each 12 μs model acquisition
Hold endpoint drift	0.295 mV median; 0.461 mV max	Endpoint-to-endpoint change across each 36 μs hold
Comparator polarity	99.67% overall; 100.0% outside ± 50 mV	Functional decision agreement around a swept threshold

The LEVEL=1 simulation at 27 °C describes the nominal acquisition, hold drift, and comparator response. The small endpoint error relative to one reference interval explains why the selected capacitor supports three-bit conversion.

5.2 Five-transistor comparison stage

I used a five-transistor OTA-like stage because the circuit needed a polarity decision around V_{ref} rather than precision linear amplification. The NMOS pair compares the held node with the selected reference, a tail device establishes current, and the BS250 mirror load creates a single-ended output. This compact stage lets the following inverter perform the final rail restoration, so the OTA does not have to behave like a high-accuracy operational amplifier.

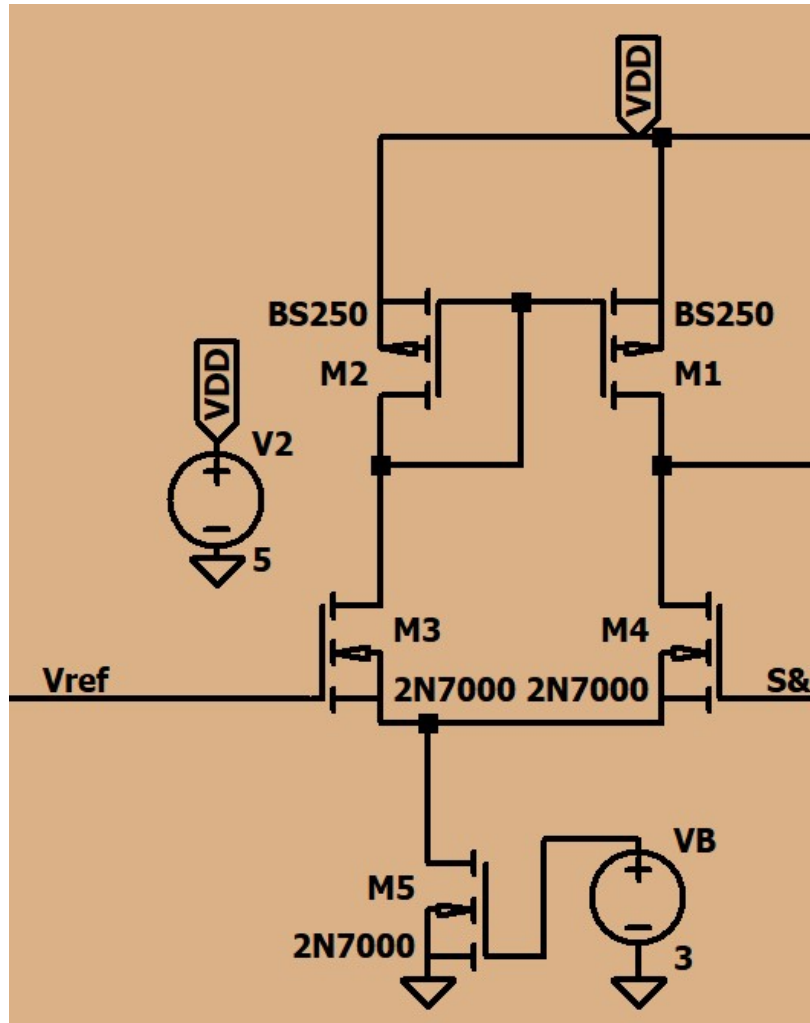


Figure 5. Five-transistor OTA-like stage used for the threshold comparison.

M3 and M4 form the differential pair, M5 sets the tail current, and M1-M2 form the PMOS mirror load. I selected the structure because it uses the available devices and turns the sign of $V(S\&H) - V_{ref}$ into a useful single-ended movement. The choice trades precision specifications for simplicity and is appropriate only because a regeneration stage follows it.

5.3 CMOS regeneration

The OTA output can move slowly and may not use the complete supply swing near a threshold crossing. I therefore followed it with a BS250/2N7000 CMOS inverter. This separates two tasks: the OTA senses which input is larger, and the inverter creates the steep, near-rail decision that the digital controller can sample.

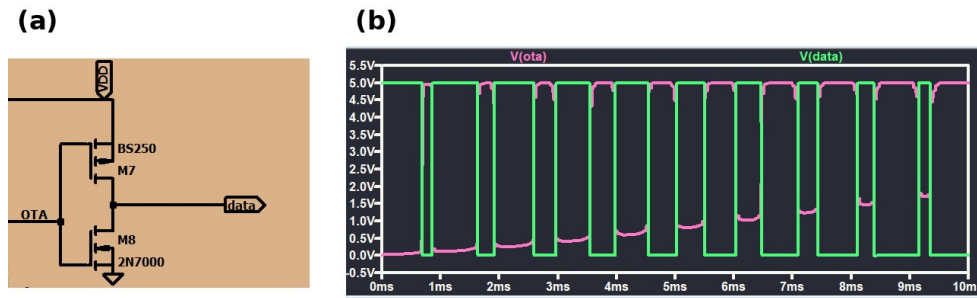


Figure 6. Discrete CMOS regeneration stage: (a) inverter topology and (b) nominal regenerated output.

The same BS250/2N7000 device families were used to keep the comparison path discrete. Panel (b) shows the data node approaching the supply rails while the OTA node changes more gradually, which is the required regeneration behavior.

6 Reference Generation and Mixed-Voltage Interface

6.1 Outgoing-control translation and polarity

The Basys3 control signals use 0/3.3 V logic, while the breadboard analog controls operate around 5 V. I used two cascaded BS250/2N7000 CMOS inverters to translate the control voltage and preserve its logical polarity. HOLD is active high during conversion, and SEL[2:0] carries the natural-binary reference address.

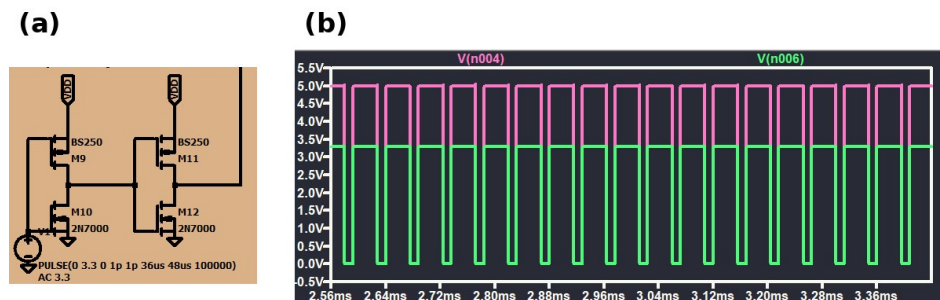


Figure 7. Control-level translation study: (a) two-inverter topology and (b) response at the 48 μ s control period.

The first inverter changes the logic level and polarity; the second restores the polarity while supplying the 5 V control swing. The simulated input and output traces show the command transferred at the conversion-control rate.

Using the same BS250 and 2N7000 devices for translation kept the control circuit consistent with the discrete analog stages. Two inversions preserve the controller's logic: a high HOLD command keeps the input sample stored during comparison, and the three selector lines present the chosen binary address to the multiplexer.

Table 7. Logical and electrical polarity across the outgoing-control boundary.

Interface layer	Signal meaning	Electrical polarity
top_adc	hold and sel[2:0]	Active-high HOLD; natural-binary selector address

Interface layer	Signal meaning	Electrical polarity
PMOD outputs	hold and sel[2:0]	Controller polarity at 3.3 V
Two-stage level converters	HOLD and SEL at 5 V	Same polarity in the analog control domain

The control circuit links the 3.3 V FPGA logic to the 5 V analog stages. Keeping the logical HOLD and selector meanings explicit makes the switch and multiplexer operation clear at both sides of that interface.

6.2 Concentrating eight references in the useful signal window

I did not distribute the references uniformly from 0 V to 5 V because the nominal signal occupies only about 2.5–4.5 V. The LTspice model used for this report has 18 k Ω above the tap chain, eight 10 k Ω sections, and 82 k Ω below it. The total is 180 k Ω , so the ladder draws about 27.78 μ A and produces 0.2778 V spacing. Concentrating the codes in this window makes each of the three bits useful for the expected input instead of spending codes near rails that the waveform does not visit.

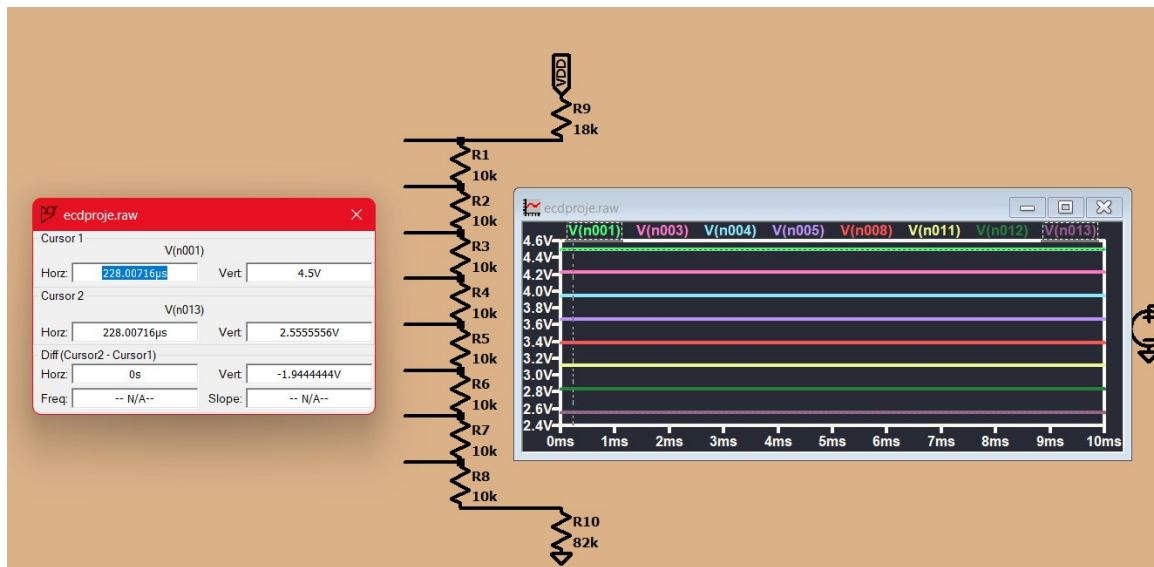


Figure 8. LTspice check of the final resistor-ladder taps.

The simulated taps are monotonic from approximately 2.556 V to 4.500 V with about 0.278 V between adjacent levels. This static result defines the useful reference window. The multiplexer selects one of these levels for each comparison in the SAR sequence.

Table 8. Reference levels produced by the final 18 k Ω + eight 10 k Ω + 82 k Ω ladder.

Selector	Nominal Vref	Role in the available range
000	2.5556 V	lowest available tap
001	2.8333 V	intermediate reference
010	3.1111 V	intermediate reference

Selector	Nominal Vref	Role in the available range
011	3.3889 V	intermediate reference
100	3.6667 V	intermediate reference
101	3.9444 V	intermediate reference
110	4.2222 V	intermediate reference
111	4.5000 V	highest available tap

The active SAR controller uses 000 while idle and visits binary-search trial boundaries from 001 through 111 during conversion. That distinction matters: the network provides eight selector voltages, while a particular three-comparison path uses only the boundaries required by the decisions already made.

6.3 Comparator return to the FPGA

The regenerated comparator decision returns on *din*, which is constrained as a 3.3 V LVCMOS input at K17. The interface therefore has one clear electrical requirement: the low and high levels at the FPGA pin must remain inside the LVCMOS33 input ranges while the comparator and control logic are switching. Each stable comparator decision is sampled by the controller to determine one bit of the ADC result.

Table 9. Comparator-return interface and use in the conversion sequence.

Return-path item	Implemented condition	Engineering role
Comparator data	Regenerated one-bit decision	Provides <i>din</i> for each SAR comparison
FPGA input	K17, LVCMOS33	Defines the accepted return-voltage range
Controller input	<i>din</i> sampled during each comparison state	Contributes one bit to the ADC result

6.4 Integrated nominal analog model and threshold behavior

The integrated LTspice transient combines the sample switch, hold capacitor, OTA-like stage, inverter, and HOLD translation. An ideal Vref sweep from 2.5 V to 4.5 V makes the comparator boundary visible across the signal window and isolates the sampling and polarity behavior from selector timing.

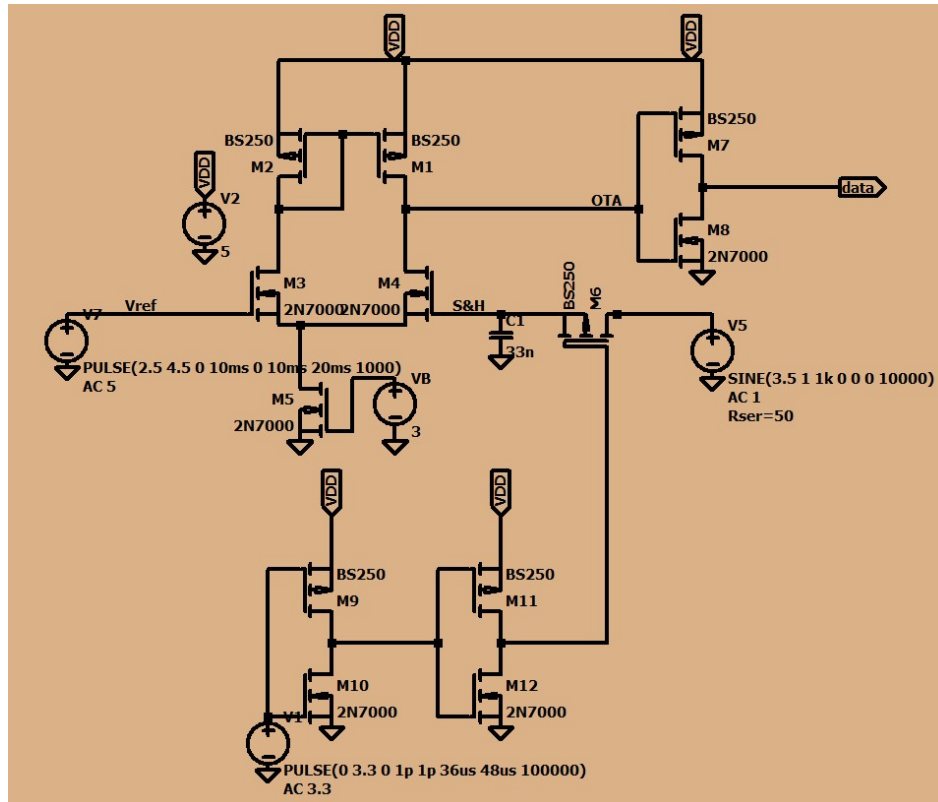


Figure 9. Integrated analog subcircuit used for the nominal transient study.

The model joins sampling, comparison, regeneration, and HOLD translation. The continuous reference ramp makes threshold polarity easy to observe; the eight-level selector sequence is described separately by the SAR controller and ladder table.

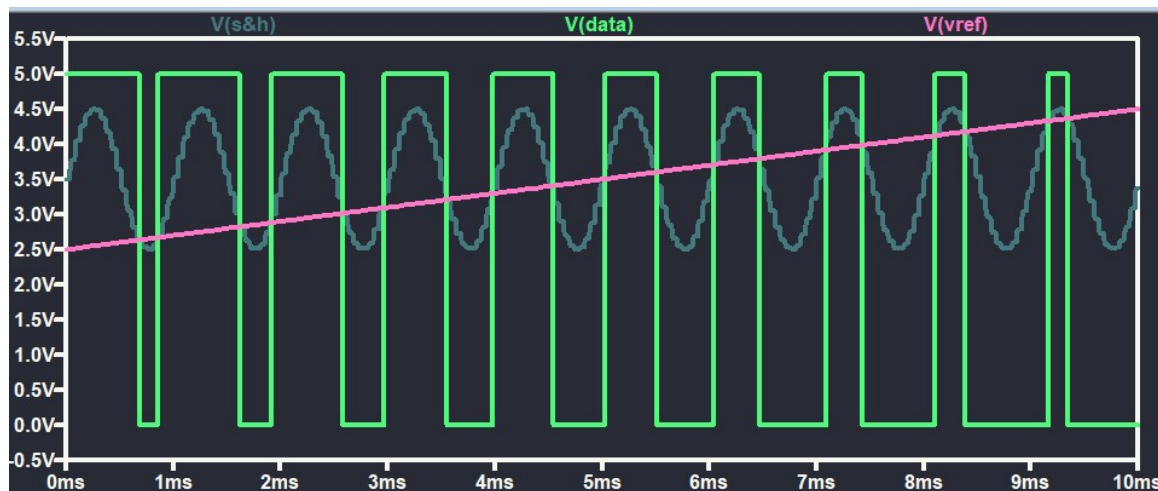


Figure 10. Nominal threshold sweep of held input, ideal reference, and regenerated data.

The data output is high when the held input exceeds the swept reference and low otherwise. Across all adaptive solver points, the output agrees with that expected polarity at 99.67%; outside a ± 50 mV band around the crossing, agreement is 100%. The finite differences cluster at transitions. The output follows the polarity needed by the successive-approximation controller.

The integrated transient shows the complete analog sequence: the input is sampled, the held value is compared with a reference, and the regenerated output supplies a digital decision.

7 Digital Conversion, Triggering, and Capture

The converter is implemented as a paced sequence rather than a parallel bank of thresholds. Inside `top_adc`, the logical selector is 000 and logical HOLD is low while the input tracks. The controller then asserts logical HOLD and performs three comparisons, selecting a midpoint for the most significant bit and refining the reference from the decisions already stored. Each state lasts three rising `dclk` edges so that the external selector and comparator have time to respond before `din` is sampled. `top_main` connects the controller outputs to the external interface.

7.1 From the planning algorithm to the active SAR controller



Figure 11. Source-grounded three-comparison sequence implemented in `top_adc`.

The controller makes the timing contract explicit. `idle` allows tracking; `bit0` tests logical selector 100 and stores the MSB; `bit1` and `bit2` construct the next logical selector from the decisions already captured. `code_valid` is asserted only after the LSB decision, so downstream logic can treat one event as one complete three-bit sample.

`counter1` reaches 199 on the 100 MHz clock before toggling `dclk`, so the internal conversion clock is 250 kHz. The four controller states each consume three rising `dclk` edges. One complete sample therefore uses twelve 4 μ s intervals, or 48 μ s.

$$f_{\text{dclk}} = \frac{100 \text{ MHz}}{2(199+1)} = 250 \text{ kHz}$$

$$T_{\text{sample}} = \frac{12}{250 \text{ kHz}} = 48 \text{ } \mu\text{s}$$

$$f_{\text{sample}} = 20.833 \text{ kSa/s}$$

Table 10. SAR conversion cadence and the logical action taken in each state.

State	Nominal duration	Control and decision
idle	3 <code>dclk</code> edges = 12 μ s state budget	logical SEL = 000; logical HOLD remains low until the transition; input tracks for approximately 8 μ s
bit0	3 <code>dclk</code> edges = 12 μ s	logical HOLD = 1; select 100; store <code>dout_b(2)</code> from <code>din</code>
bit1	3 <code>dclk</code> edges = 12 μ s	logical SEL = <code>dout_b(2)</code> & 10; store <code>dout_b(1)</code>
bit2	3 <code>dclk</code> edges = 12 μ s	logical SEL = <code>dout_b(2:1)</code> & 1; store LSB; pulse <code>code_valid</code>

7.2 Triggering from completed codes

A free-running capture can begin at a different waveform phase every time, so even a correct sequence of codes can appear to drift horizontally. I solved that user-visible problem with a trigger state machine that advances only when one conversion is complete. It first requires a code at or below 3, then detects a rising crossing into code 4 or above, waits for another low excursion, and identifies the next rising crossing as one complete period. The low/high pair gives one-code hysteresis around midscale.

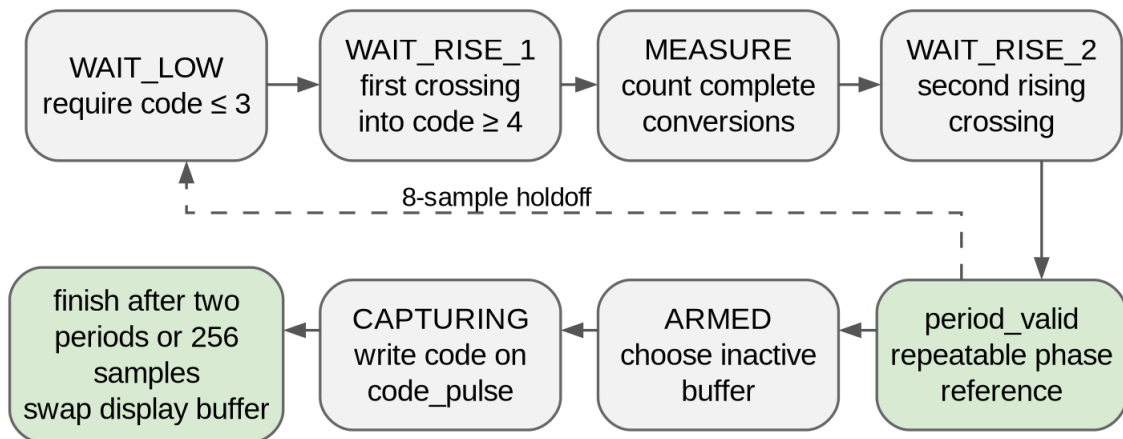


Figure 12. Implemented period detection followed by inactive-buffer capture.

period_valid is the alignment event for capture rather than a frequency measurement. When the synchronized pulse arrives, the controller chooses the inactive display buffer and writes one code on each code pulse. Capture stops after two later period events or 256 samples, and only then is the completed buffer exposed to the renderer. The visible trace therefore remains stable while the next record is collected.

Table 11. Trigger and capture conditions implemented in the VHDL.

Stage	Condition	Reason for the condition
WAIT_LOW	completed code ≤ 3	Arm only after the signal is clearly below the midscale boundary
WAIT_RISE_1	prior code < 4 ; current code ≥ 4	Establish the first repeatable rising crossing
MEASURE	count completed conversions until code ≤ 3	Require a real low excursion before accepting another rise
WAIT_RISE_2	next crossing: prior code < 4 ; current code ≥ 4	Mark one full low-to-high waveform cycle
HOLDOFF	8 completed samples	Reduce immediate re-triggering near the crossing
CAPTURING	code_pulse; stop at two periods or 256	Write one complete sample per event and bound memory use

7.3 Digital module hierarchy

top_main connects the Clocking Wizard, top_adc, and vga_adc_scope_3bit. The Clocking Wizard derives the 25 MHz pixel clock and its locked output holds the VGA logic in reset until the clock is stable. top_adc supplies HOLD, the reference address, the three-bit result, and the trigger event. top_main connects these signals to the external interface and the capture module. The scope block synchronizes conversion events into the 100 MHz capture process and then passes completed-display metadata into the pixel domain.

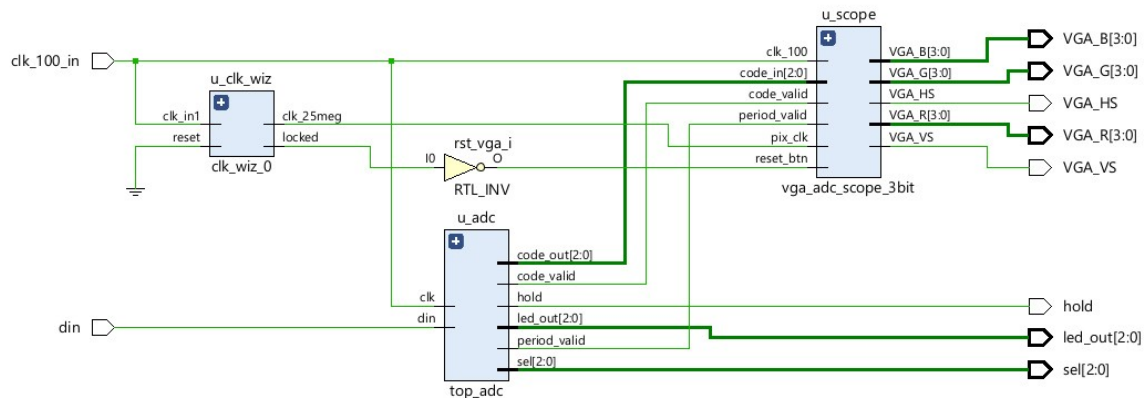


Figure 13. Vivado elaboration of the top-level digital hierarchy.

The view shows the implemented hierarchy: one clock generator, one conversion/trigger controller, and one capture/VGA module. Appendix C lists the corresponding design files.

8 VGA Rendering and User-Visible Behavior

The project result had to be visible on a normal monitor, so I treated display stability as part of the engineering problem. The renderer uses a 25 MHz clock and the standard 800×525 total raster associated with a 640×480 visible area. The resulting refresh rate is about 59.52 Hz. I draw a white background, black axes, and a three-pixel-thick black code trace inside a 540-pixel-wide by 281-pixel-high plot region.

$$\text{frame rate} = \frac{25 \text{ MHz}}{800 \times 525} \approx 59.52 \text{ Hz}$$

Table 12. Implemented VGA timing, plot geometry, and memory organization.

Quantity	Implemented value	How I used it
Visible raster	640×480	Monitor image area
Horizontal total	800 pixels = $640 + 16 + 96 + 48$	Visible area, front porch, sync, and back porch
Vertical total	525 lines = $480 + 10 + 2 + 33$	Visible area, front porch, sync, and back porch
Pixel clock	25 MHz	Clock Wizard output used by timing and rendering

Quantity	Implemented value	How I used it
Plot box	x = 50...589; y = 100...380	540 display columns and eight vertical code levels
Capture memory	2 × 256 × 3-bit arrays	One visible buffer and one inactive write buffer
Vertical code step	40 pixels	Map codes 0...7 into eight separated horizontal levels

8.1 Stretching a variable capture across the plot

The number of stored samples changes with input frequency because capture follows two detected periods. I therefore stretch the completed capture across the fixed 540-pixel plot width. For each visible x coordinate, the renderer computes an integer sample index from the horizontal position and the stored sample count, clamps the count to 1...256, reads the selected buffer, and maps the three-bit value to one of eight y levels.

$$\text{sample_index} = \frac{(x - X_START)(n_samples - 1)}{PLOT_W - 1}$$

I chose this direct mapping because it makes every completed capture use the available screen width without a preprocessing pass. The design draws horizontal code segments only. That choice makes the quantization levels easy to read, but it also explains why square-wave and sawtooth transitions appear as separated plateaus rather than connected vertical edges in the photographs.

8.2 Cross-domain transfer

code_valid and period_valid are synchronized into the 100 MHz capture domain and edge-detected so that a stretched synchronized level produces only one event. The three-bit code is passed through a two-stage register chain and sampled when code_pulse is accepted. Display selector and sample count are then passed through two registers in the pixel domain. The renderer uses the completed buffer and its sample count to place the captured waveform across the plot width. This keeps the visible record available while the other buffer collects new samples.

9 Hardware Construction and Integration

I assembled the analog chain and reference network on solderless breadboards and connected the control and decision signals through the Basys3 PMOD headers. The FPGA, analog circuit, and VGA output remained physically separate so that HOLD, the held node, selected references, and the comparator output could be probed during bring-up.

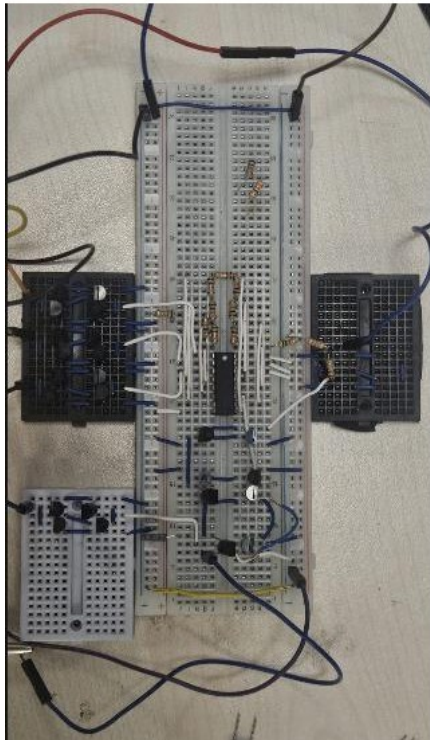
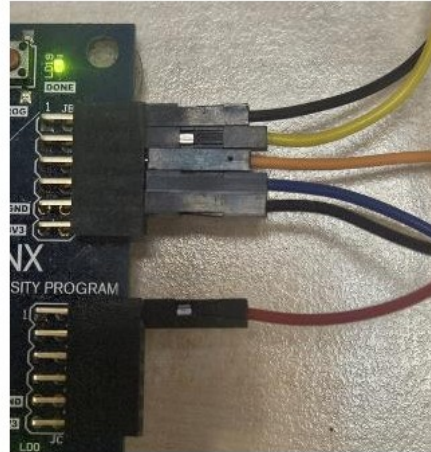
(a)**(b)**

Figure 14. Hardware construction details: (a) discrete analog breadboard and (b) Basys3 PMOD connection.

The breadboard kept individual analog nodes accessible while the adjacent JB pins grouped the four outgoing controls for probing. This arrangement made the held node, selector lines, comparator return, and FPGA pins easy to inspect during integration. The four PMOD controls carry HOLD and the three-bit reference address; the table below records their pin connections.

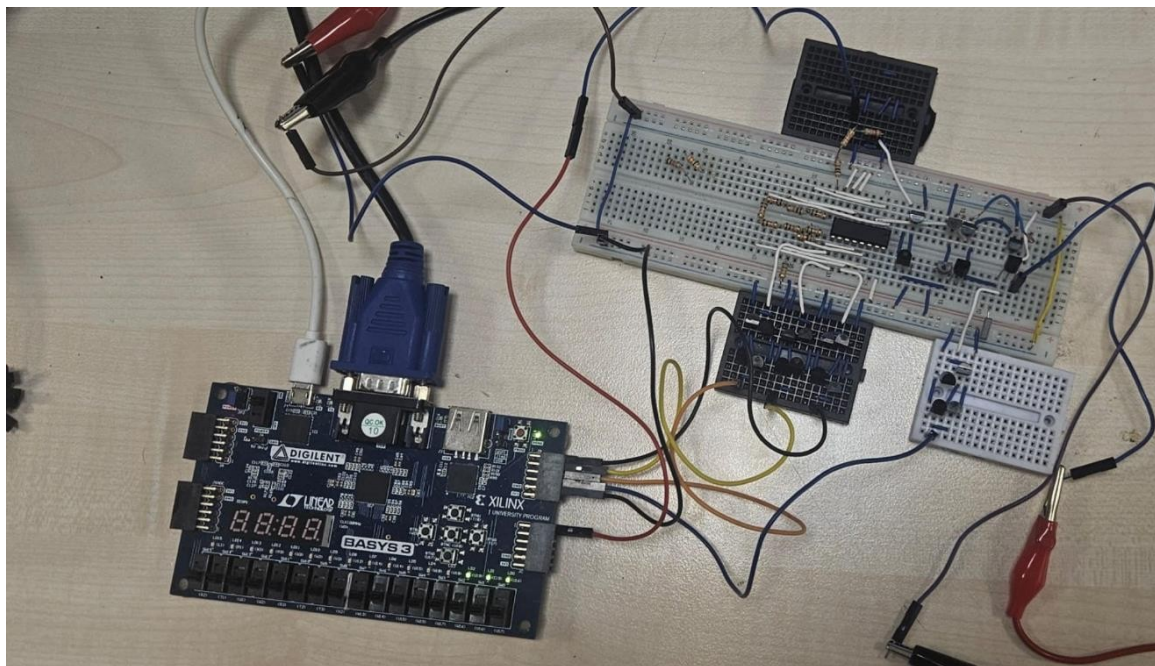


Figure 15. Integrated Basys3, analog breadboards, PMOD wiring, and VGA cable.

The photograph shows the complete physical setup operating as one instrument: Basys3, analog breadboards, PMOD wiring, oscilloscope access, and VGA output. This arrangement connects the circuit-level tests with the waveform demonstrations in Chapters 10 and 11.

Table 13. Principal Basys3 interfaces used by the integrated system.

Function	Port	Basys3 pin / interface	Purpose during bring-up
System clock	clk_100_in	W5	100 MHz controller and Clocking Wizard source
Reference select	sel[0], sel[1], sel[2]	A14, A16, B15 / JB1–JB3	Natural-binary address through the two-stage level converters
Sample control	hold	B16 / JB4	Active-high HOLD through the two-stage level converter
Comparator return	din	K17 / JC1 comment	Return the one-bit analog decision
Latest code	led_out[0..2]	U16, E19, U19	Board-level observation of the three-bit result
Display	VGA RGB, HS, VS	Basys3 VGA connector	Drive the 640 × 480 waveform view

10 Testing and Bench Results

I started bench testing at the analog boundary. I checked the held node first because each comparator decision depends on a stable sample. I then compared the control activity with the held waveform before exercising the complete system with sine, square, triangle, and sawtooth inputs. Appendix B contains the additional lower-frequency, reference-voltage, spectral, and high-frequency tests.

Table 14. Bench tests and what each one showed.

Test	What I checked	What I observed
Held-node scope view	Does the physical stage preserve samples?	Held staircase waveform with oscilloscope averaging set to 128
Hold/control view	Is control activity present at the held waveform?	Control activity displayed alongside the held-waveform reference
1 kHz waveform family	Does the complete system distinguish shape?	Recognizable VGA sine, square, triangle, and sawtooth outputs

Test	What I checked	What I observed
500 Hz and 200 Hz sine	Does capture adapt to longer periods?	More samples per period and longer on-screen spans; Appendix B
MUX/reference spot-check	Is a midrange voltage present on hardware?	3.542 V reading on the breadboard reference network
Exploratory FFT	Does the held signal show a dominant component?	46 dB cursor separation in the displayed spectrum
12.95 kHz limit	What happens beyond the nominal bandwidth?	Sparse, distorted output with changed capacitor and clock-divider settings

10.1 Physical sample-and-hold observations

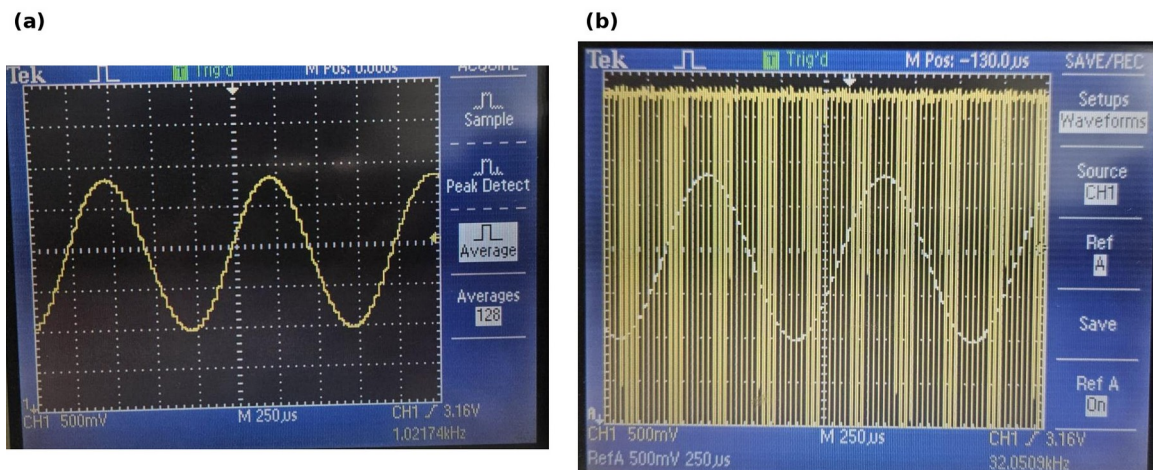


Figure 16. Breadboard sample-and-hold observations: (a) averaged held waveform and (b) live control activity with stored Ref A.

Panel (a) shows the staircase expected when the 1 kHz input is periodically sampled, providing the physical link between the LTspice result and the complete converter. Averaging makes the levels easy to distinguish. Panel (b) places the control activity and held-waveform reference on the same time scale and shows their repeated relationship during operation.

11 Laboratory Presentation and VGA Results

After the held node was checked, I exercised the FPGA and breadboard as one instrument. The four 1 kHz photographs below form the representative end-to-end result because each waveform exposes a different aspect of conversion and rendering. Together they show acquisition, three-bit conversion, triggering, buffered capture, and VGA display across four input shapes.

11.1 One-kilohertz waveform family

At the nominal 20.833 kSa/s rate, a 1 kHz input produces about 20.8 samples per period and approximately 41.7 samples across the intended two-period capture. That count is enough to show the main shape while keeping the three-bit staircase visible. I used four waveform families because each stresses a different part of the system: the sine exercises smooth threshold progression, the square wave exposes transition rendering, the triangle checks near-linear code movement, and the sawtooth combines a ramp with an abrupt reset.

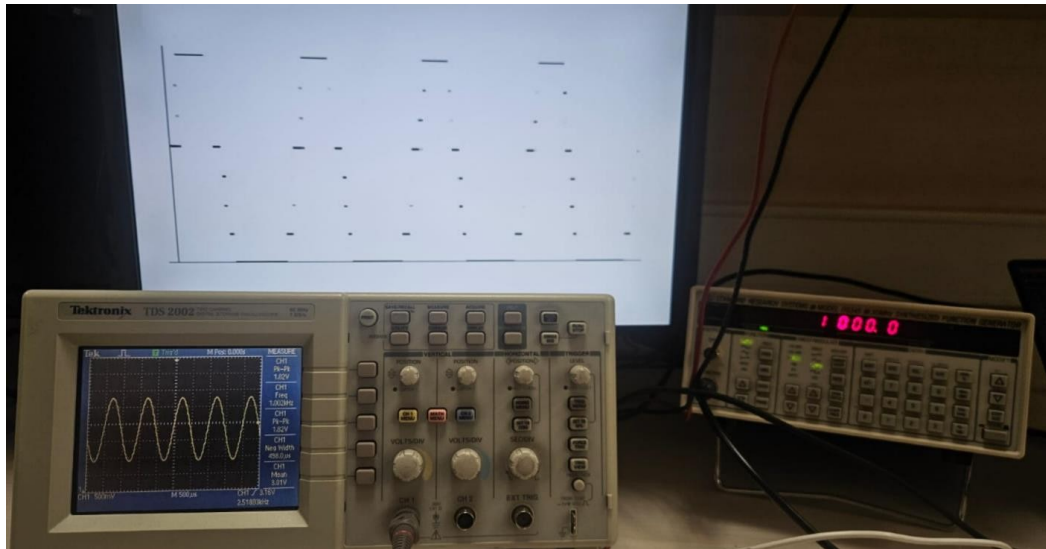


Figure 17. End-to-end 1 kHz sine-wave demonstration.

The oscilloscope shows the applied sine while the VGA monitor shows a stable discrete approximation from the external analog front end, three-comparison controller, trigger, capture memory, and renderer. The result demonstrates the complete signal path and makes the expected three-bit staircase visible.

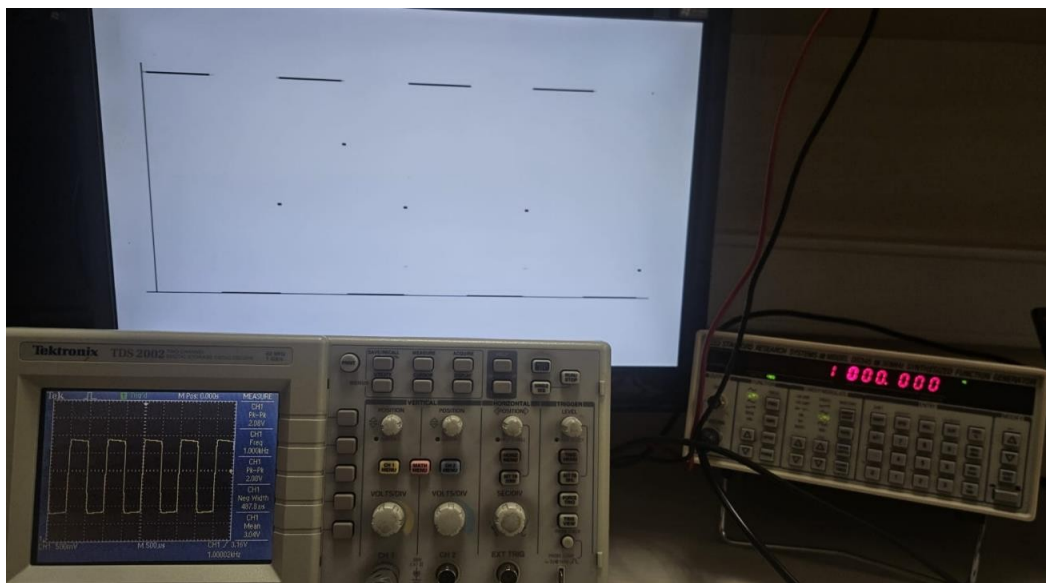


Figure 18. End-to-end 1 kHz square-wave demonstration.

The system distinguishes the high and low plateaus and holds the trace at a repeatable phase. The vertical transition is not continuous because the renderer places horizontal segments at

sampled code levels and does not connect adjacent y values. This is a display choice rather than an intermediate-voltage measurement.

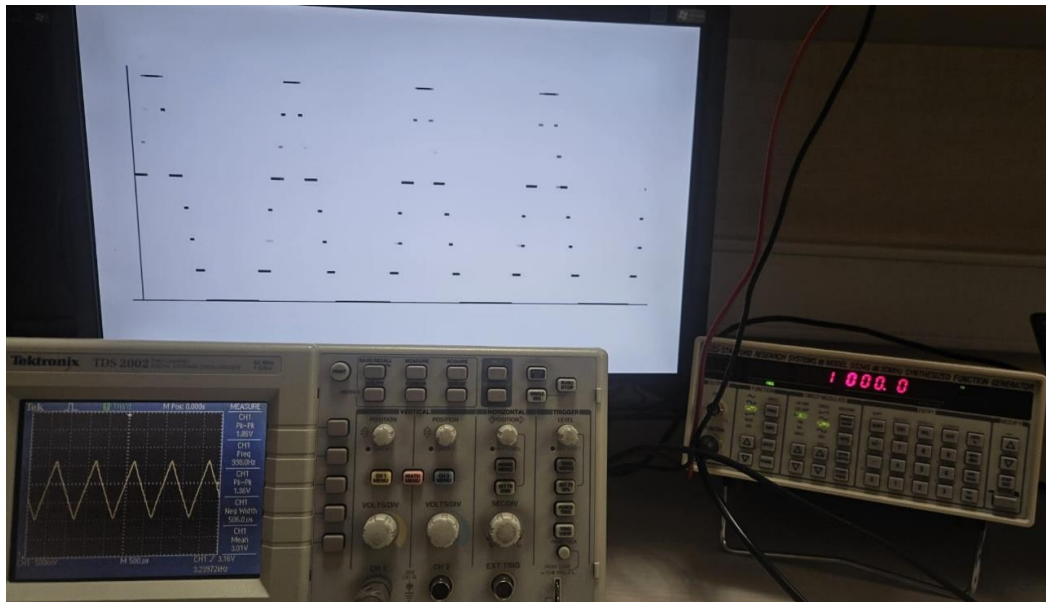


Figure 19. End-to-end 1 kHz triangle-wave demonstration.

The displayed code progresses up and down through the available levels in the expected order, so the triangle test is a useful monotonicity check across repeated thresholds. The low three-bit resolution makes the staircase pronounced, which is expected.

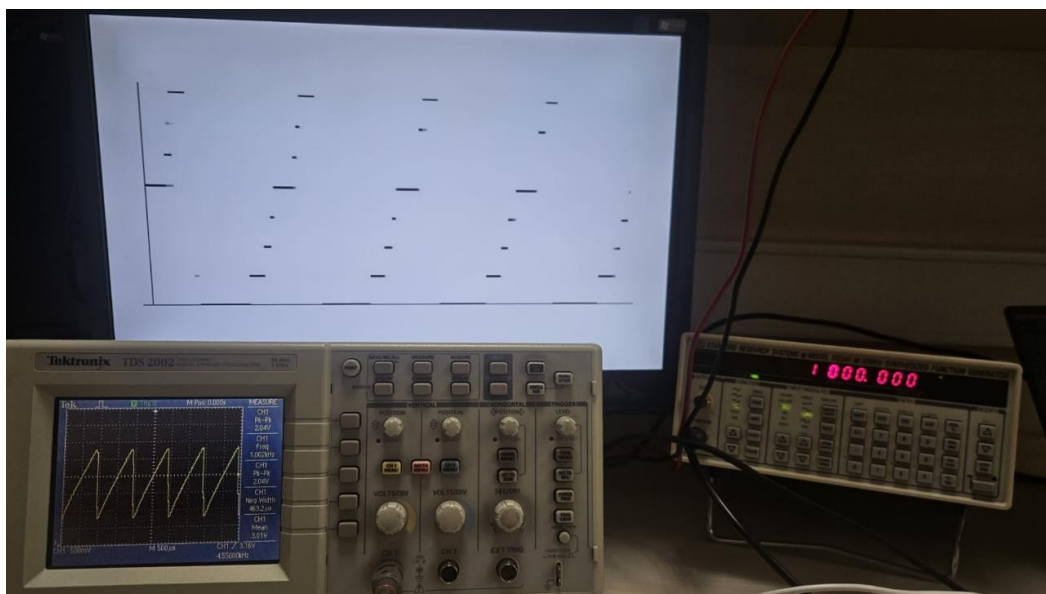


Figure 20. End-to-end 1 kHz sawtooth-wave demonstration.

The gradual ramp and abrupt reset exercise both ordered threshold movement and a discontinuity. The rising codes are recognizable, while the reset again appears as separated horizontal segments because the renderer omits vertical connectors. This test helped me distinguish a display-style limitation from the analog conversion behavior.

11.2 Frequency variation within the capture memory

The two-period capture makes the expected sample count easy to estimate. At 500 Hz it is about 83 samples for two periods. At 200 Hz it is about 208 samples, still below the 256-sample limit. This explains why the same renderer can show progressively longer discrete periods without changing the memory or display geometry.

Table 15. Expected sample occupancy for the demonstrated sine-wave frequencies.

Input frequency	Samples per period at 20.833 kSa/s	Approximate two-period samples	Buffer implication
1 kHz	20.8	41.7	Comfortably within 256
500 Hz	41.7	83.3	Comfortably within 256
200 Hz	104.2	208.3	Within 256 with less margin
Below ≈ 162.8 Hz	>128	>256	Two periods would reach the buffer limit

11.3 Additional bench observations

Appendix B adds the 500 Hz and 200 Hz sine demonstrations, a 3.542 V reference-network check, an exploratory FFT screen, and a 12.950 kHz limit case. The lower-frequency traces show the increased number of samples per period, the meter records a midrange reference voltage, and the FFT view shows a dominant component with 46.0 dB cursor separation. The additional high-frequency test used a smaller hold capacitor and an adjusted conversion-clock divider.

Table 16. Summary of the completed laboratory demonstrations.

Test	Observed result	Engineering interpretation
1 kHz waveform tests	Sine, square, triangle, and sawtooth displayed on VGA	Complete breadboard acquisition and FPGA display path
500 Hz and 200 Hz sine	Stable traces with more samples per period	Capture length follows the input period
Sample-and-hold	Staircase waveform on the oscilloscope	The capacitor holds successive input samples
Reference network	3.542 V meter reading	Midrange reference within the conversion window
FFT and 12.950 kHz input	Dominant spectral component; distorted high-frequency VGA trace	Spectral observation and an additional high-frequency experiment

12 FPGA Implementation

Basys3 performs the digital part of the converter and drives the VGA monitor. I described the conversion sequence, trigger, capture buffers, and display logic in VHDL. The top-level module connects `top_adc`, `vga_adc_scope_3bit`, and the Clocking Wizard, while the XDC assigns the clock, PMOD signals, LEDs, and VGA connector pins.

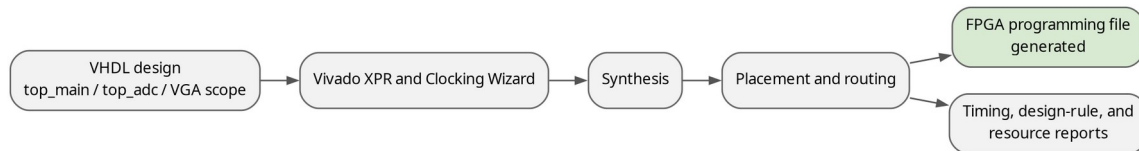


Figure 21. VHDL implementation flow for the Basys3.

Vivado translates the VHDL into FPGA logic and generates the configuration used to program Basys3. Once configured, the FPGA repeatedly controls the breadboard converter and renders the captured samples without a host computer.

Table 17. FPGA functions used in the completed laboratory system.

Function	Implementation	Role during the demonstration
Conversion control	<code>top_adc</code> state machine	Drive HOLD and reference selection; sample comparator decisions
Waveform trigger	Low/high threshold detection	Align each capture to a rising waveform crossing
Capture storage	Two 256×3 -bit buffers	Collect a new record while displaying the completed record
VGA clock	25 MHz from <code>clk_wiz_0</code>	Set the pixel scan rate
VGA display	<code>vga_adc_scope_3bit</code>	Map ADC codes to the eight vertical levels
Top-level integration	<code>top_main</code> and <code>xdc.xdc</code>	Connect the analog interface and monitor to Basys3

12.1 Conversion and display timing

The 100 MHz oscillator supplies the FPGA system clock. A divided 250 kHz conversion clock gives $4 \mu\text{s}$ between controller edges, and twelve edges complete the $48 \mu\text{s}$ conversion cycle. The independent 25 MHz pixel clock scans the VGA image continuously. Capture buffers connect these rates: completed ADC samples enter one buffer while the renderer reads the other.

This division let the converter follow its analog settling schedule while the VGA interface maintained a continuous raster. The laboratory display showed the acquired waveform at a stable phase as the generator changed input shape and frequency.

13 Discussion and Conclusion

13.1 Relationship between the circuit and displayed waveform

The laboratory results connect the analog and digital parts of the project. The sample-and-hold keeps one input value available during the three comparisons. The comparator and selected ladder references turn that value into a three-bit code. Triggered capture then places successive records at a repeatable phase, and the VGA renderer makes the eight code levels visible.

The oscilloscope traces show the held staircase and its relationship to the control signal. The signal-generator, oscilloscope, and monitor photographs demonstrate sine, square, triangle, and sawtooth conversion through the assembled breadboard and Basys3. The lower-frequency tests also show that capture follows the input period: the 200 Hz waveform uses more of the buffer than the 1 kHz waveform.

13.2 Operating characteristics and engineering lessons

Three-bit quantization gives eight amplitude levels, so the VGA trace is deliberately coarse. The square-wave edges and sawtooth reset appear as separated horizontal segments because the renderer draws sampled levels without vertical connectors. These features explain the appearance of the completed display.

The 48 μ s conversion period gives about 20.8 samples per cycle at 1 kHz. At 500 Hz and 200 Hz, the larger sample counts make the discrete shape easier to follow. An additional 12.950 kHz experiment used a smaller hold capacitor and an adjusted conversion-clock divider. Its sparse, distorted display illustrates the interaction between analog settling, conversion cadence, and the input frequency.

Using Basys3 brought the conversion sequence and display into the same VHDL design. The project required the analog switch timing, reference selection, comparator polarity, and display capture to agree. Building and testing these functions together on the breadboard made their interaction visible at the oscilloscope and on the monitor.

13.3 Conclusion

The project achieved its objective: I constructed and demonstrated a complete three-bit ADC on a breadboard. Basys3 implemented the digital control in place of the proposed Arduino and generated a VGA display of the sampled waveform. The completed signal path includes the MOSFET sample-and-hold, comparison and regeneration stages, reference multiplexer, successive-approximation controller, waveform trigger, and display buffers.

The laboratory presentation demonstrated the system with a signal generator and oscilloscope. Sine, square, triangle, and sawtooth inputs produced recognizable VGA traces, and the 500 Hz and 200 Hz tests demonstrated capture over longer periods. The schematics and LTspice results explain the circuit behavior, while the breadboard and instrument photographs document the functioning system.

The work combined transistor-level analog design with FPGA control and real-time waveform display. The result is a working breadboard ADC and VGA instrument that fulfills the assignment through the Basys3 implementation.

14 References

- [1] K. Gürbüz, EEE313 Term Project: VGA Interfaced 3-bit ADC with Basys3, DOCX/PDF project report, 4 January 2026.
- [2] K. Gürbüz, top_main.vhd, top_adc.vhd, and vga_adc_scope_3bits.vhd, project VHDL modules in new/.
- [3] K. Gürbüz, ecdproje.asc and ecdproje.raw, LTspice analog-front-end model and transient data.
- [4] 2N7000.txt and BS250.txt, MOSFET model files used by the project.
- [5] xdc.xdc, Basys3 clock, PMOD, LED, comparator-return, and VGA pin constraints.
- [6] Vivado project ECD_PROJE.xpr, Clocking Wizard configuration, and top-level elaborated design.
- [7] Project analog schematic, topology checks, and interface design notes.
- [8] Project photographs and screenshots organized under documentation/evidence and used throughout this report.

Appendix A Supplementary Design, Simulation, and Source Reference

This appendix collects the planning diagrams and additional simulation views that support the project, then closes with the active source files and external interface. The main chapters use the VHDL and LTspice data to describe the implemented behavior and numerical results.

A.1 Control and architecture diagrams

The state and architecture drawings explain the binary-search, trigger, and system organization. The implementation chapters then add the exact VHDL state names, timing, signal polarity, and module boundaries.

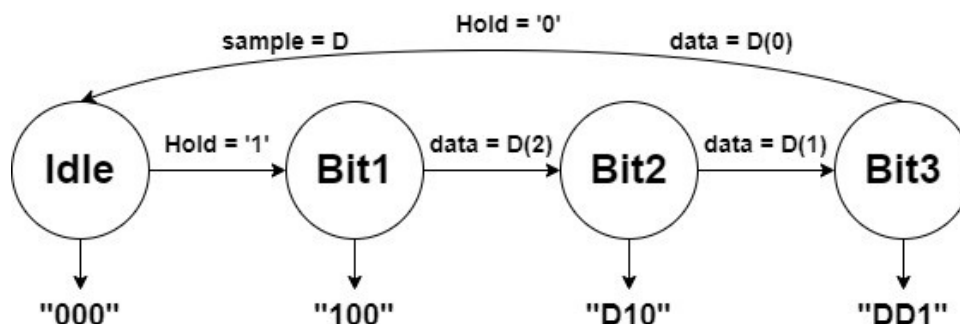


Figure 22. Sampling-algorithm state diagram.

The diagram captures the binary-search idea: hold one input, test the midpoint, and refine the reference from each decision. Chapter 7 describes the three-edge state timing and the `code_valid` event that marks a completed conversion.

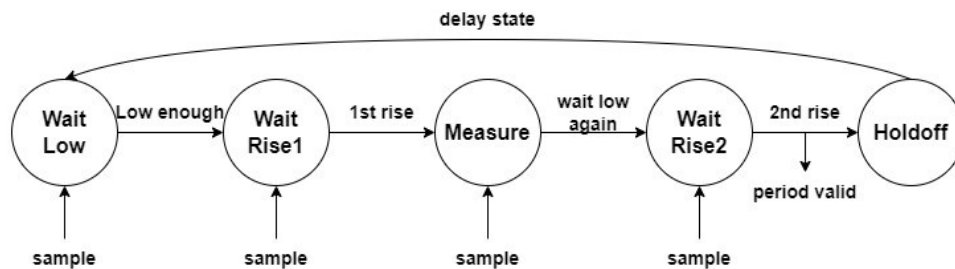


Figure 23. Trigger-state diagram.

The diagram records the need for a repeatable rising crossing. The VHDL gates state changes with completed samples, uses explicit low/high thresholds, requires a second low excursion, and applies an eight-sample holdoff.

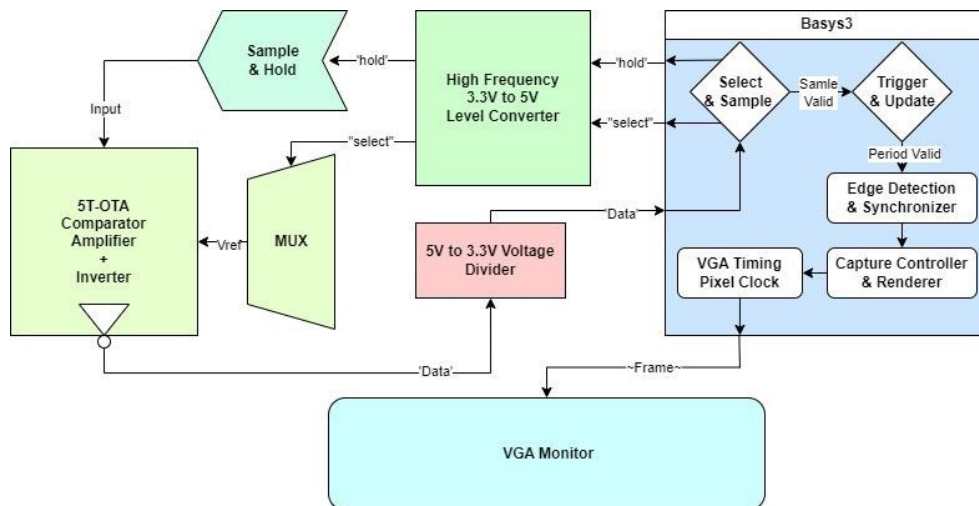


Figure 24. Project-level architecture concept.

The concept separates analog conversion from FPGA capture and display. The architecture in Chapter 4 expands the signal directions, control-level translation, module connections, capture buffers, and pixel-clock operation.

A.2 Additional simulation and reference views

These screens show the sample-and-hold transient, comparator AC response, and reference-assignment development. The RAW-derived plot and final ladder table in the main chapters add the quantitative values used in the design.

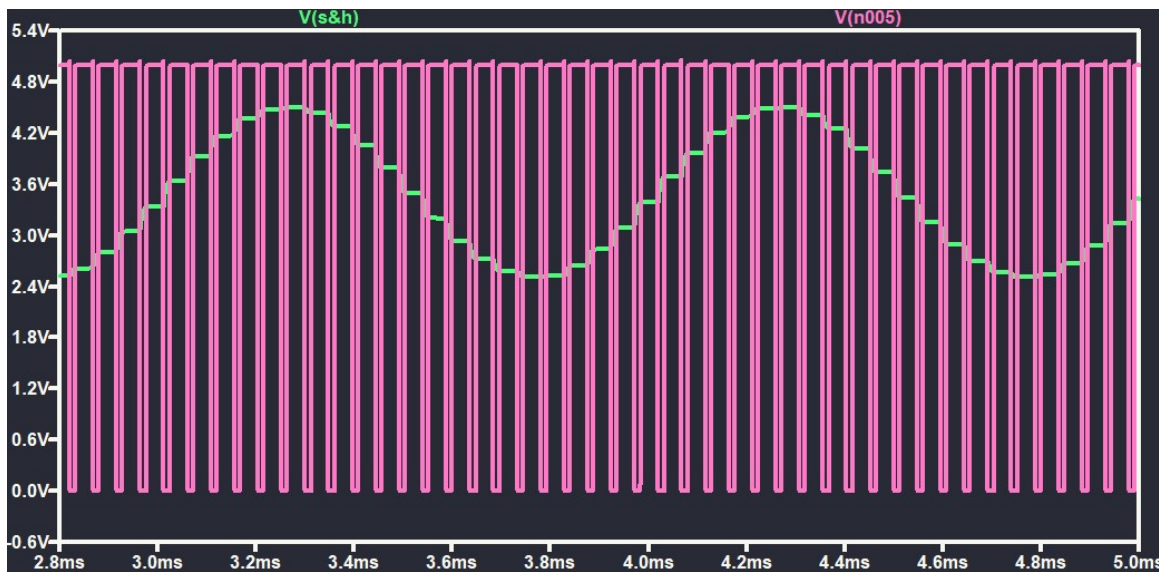


Figure 25. Sample-and-hold transient.

The held node follows the 1 kHz input during acquisition and forms a staircase during hold. The RAW analysis in Chapter 5 quantifies the endpoint error and drift visible in this waveform.

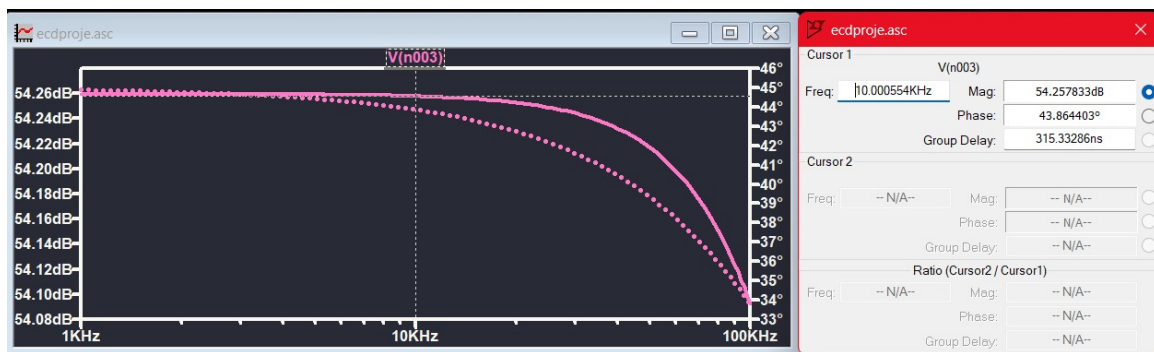


Figure 26. AC screening of the OTA-like node.

The cursor records 54.2578 dB at approximately 10.0006 kHz on V(n003). The view shows the frequency response of the plotted node. The AC plot describes the comparison-stage response over the displayed frequency range.

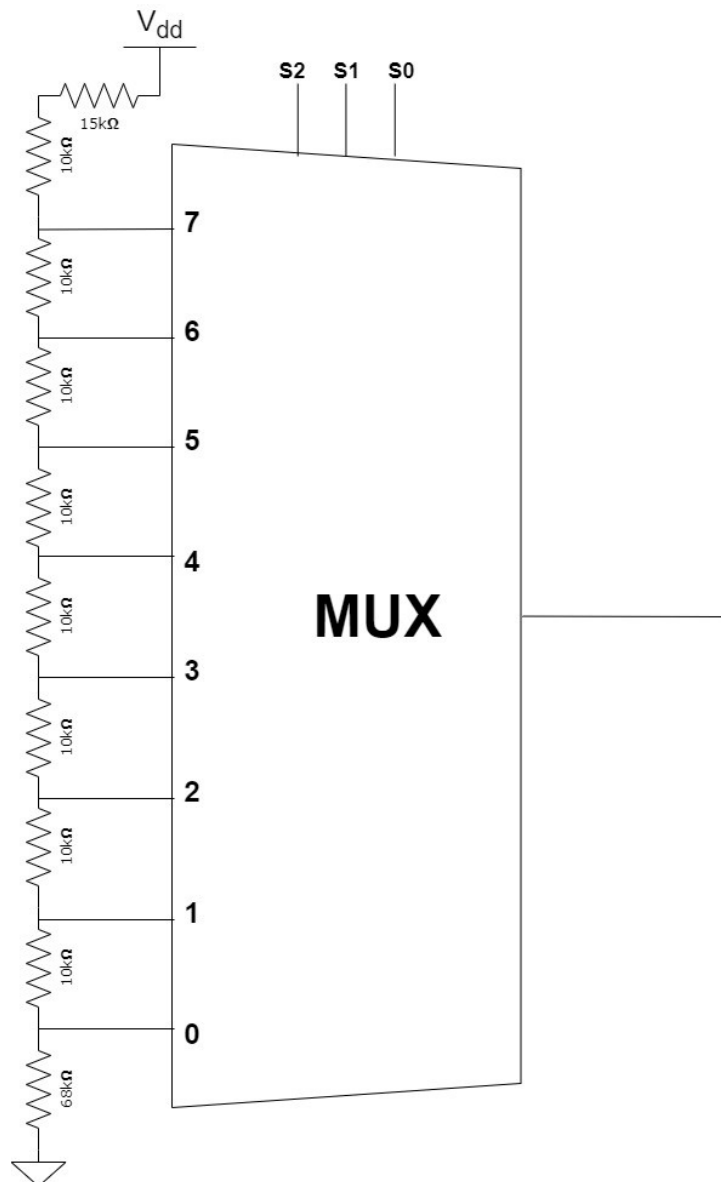


Figure 27. MUX/reference assignment drawing.

The drawing explains the selector-to-reference ordering. Chapter 6 uses the 18 k Ω /82 k Ω end resistors from ecdproj.asc to calculate the 2.556 V to 4.500 V nominal tap range.

A.3 VHDL modules and interface reference

The synthesis path uses the three VHDL files below together with clk_wiz_0 and xdc.xdc. The table identifies each module's role and its principal signals.

Table 18. VHDL module map.

File or block	Responsibility	Principal signals
top_main.vhd	Connect clock, ADC/trigger, and VGA blocks	clk_100_in, din, hold, sel, led_out, VGA

File or block	Responsibility	Principal signals
top_adc.vhd	Generate cadence, run the three comparisons, and detect a period	dclk, logical HOLD/SEL, code_out, code_valid, period_valid
vga_adc_scope_3bits.vhd	Synchronize events, capture into two buffers, and render the trace	code_in, code_valid, period_valid, RGB, HS, VS
clk_wiz_0	Derive the 25 MHz pixel clock and lock indication	clk_100_in, clk_25meg, locked
xdc.xdc	Map the active ports to Basys3 clock, PMOD, LED, comparator, and VGA pins	W5, JB pins, K17, VGA connector

Table 19. Principal Basys3 pin assignments and control signals.

Function	Port	Pin(s)	I/O standard
100 MHz clock	clk_100_in	W5	LVC MOS33
Comparator return	din	K17	LVC MOS33
Reference select	sel[0], sel[1], sel[2]	A14, A16, B15	LVC MOS33; natural-binary address
Sample control	hold	B16	LVC MOS33; active-high HOLD
Debug LEDs	led_out[0], [1], [2]	U16, E19, U19	LVC MOS33
VGA sync	VGA_HS, VGA_VS	P19, R19	LVC MOS33
VGA color	VGA_R/G/B[3:0]	Basys3 VGA pin group	LVC MOS33

Appendix B Additional Laboratory Results

The laboratory presentation also included lower-frequency sine inputs, a reference-voltage check, an oscilloscope FFT view, and a high-frequency test. These results complement the 1 kHz waveform demonstrations in Chapter 11.

B.1 Lower-frequency capture observations

At the nominal 20.833 kSa/s rate, two periods occupy about 83 samples at 500 Hz and 208 samples at 200 Hz, both within the 256-sample buffer. The two demonstrations therefore test the same period-based capture rule at longer waveform periods.

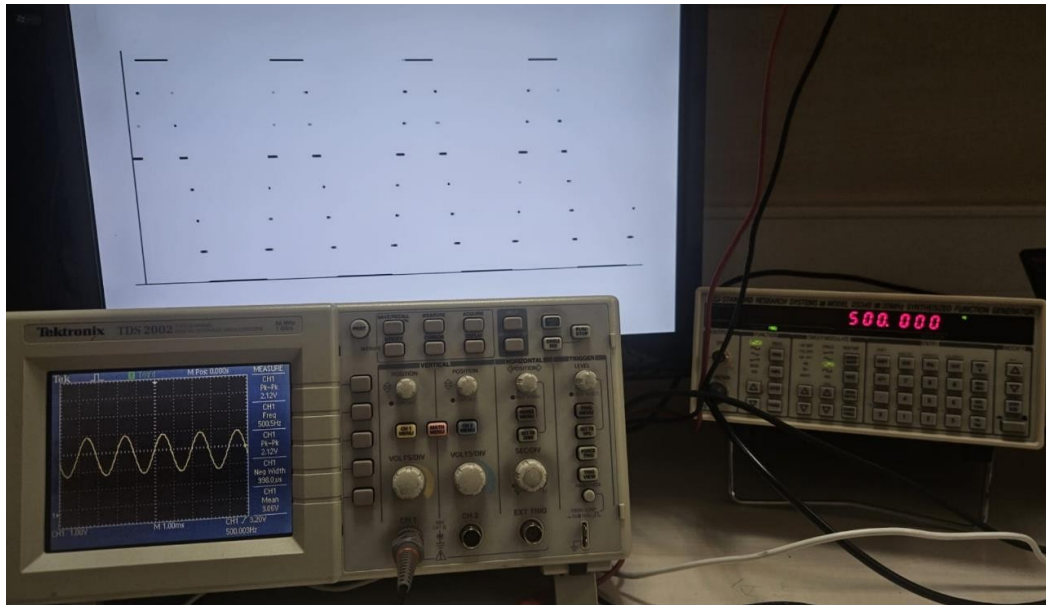


Figure 28. 500 Hz sine-wave demonstration.

The lower frequency produces roughly twice as many samples per period as the 1 kHz case. The stable trace shows the period-based capture adapting without a hard-coded waveform length.

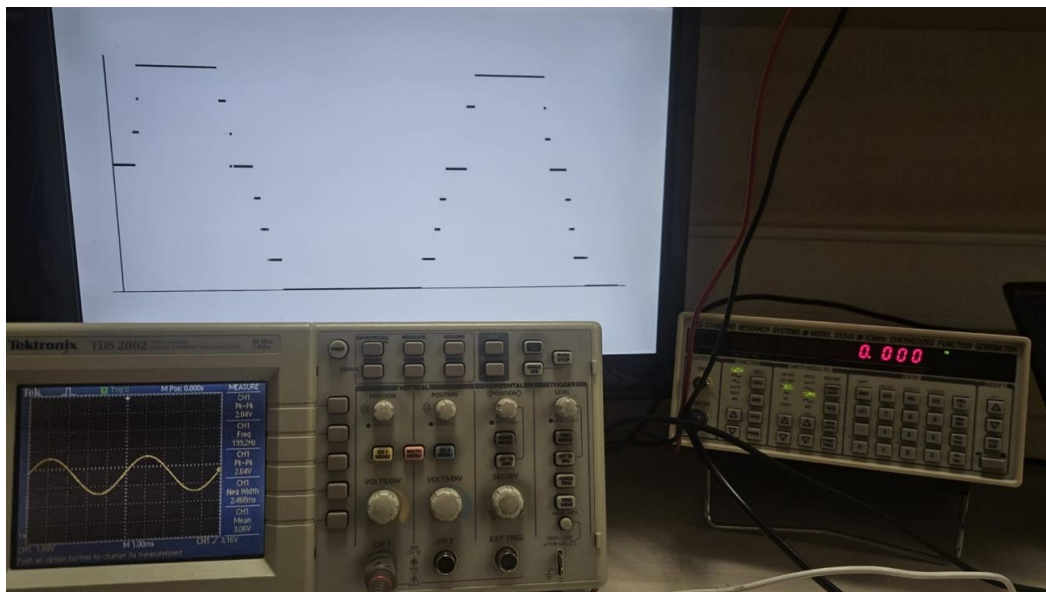


Figure 29. 200 Hz sine-wave demonstration.

The longer captured period occupies about 208 samples for two cycles, which remains below the 256-sample limit and uses most of the available capture memory.

B.2 Reference-network and spectral spot checks

These views add a 3.542 V reference-network reading and an exploratory FFT display to the waveform tests. The meter shows the reference-network operating point, and the oscilloscope FFT shows the held waveform's dominant spectral component.

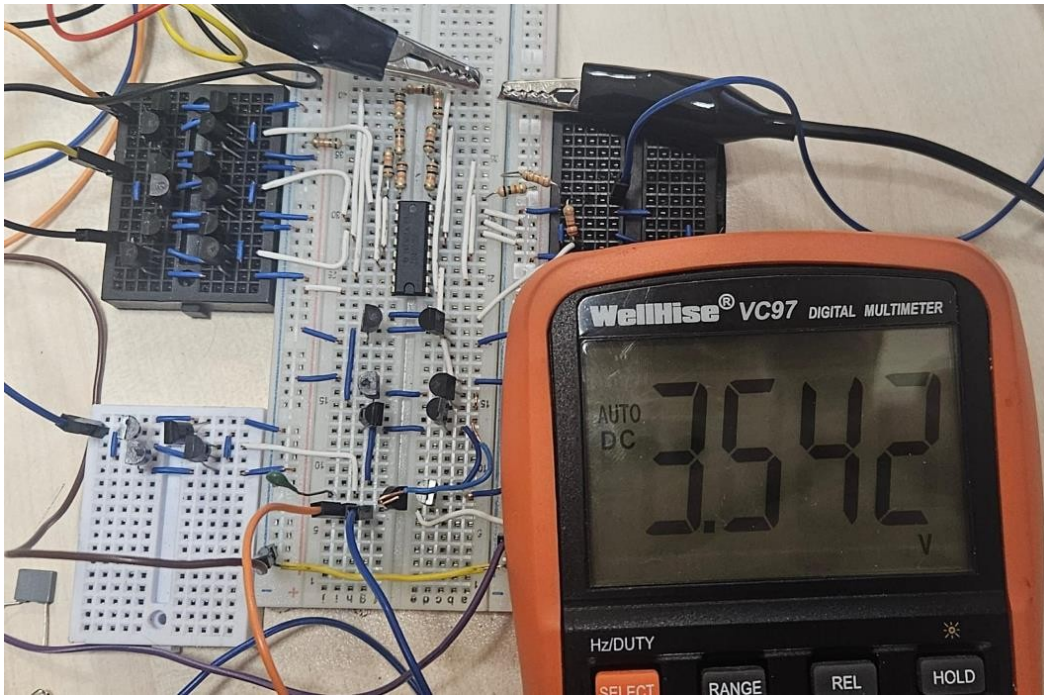


Figure 30. Single-point voltage check associated with the reference network.

The reading shows a midrange voltage on the constructed reference network. This operating point lies within the reference window used by the converter.

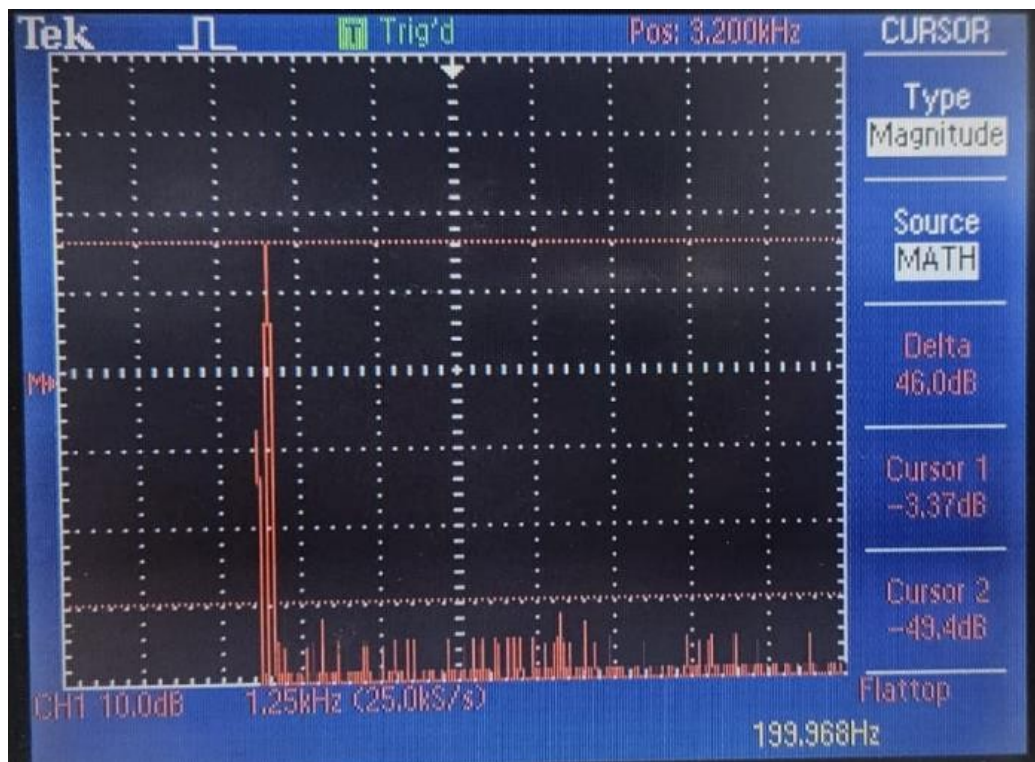


Figure 31. Exploratory FFT screen of the held waveform.

The screen shows a dominant component and a 46.0 dB cursor separation. The Flattop window and cursor positions shown on the oscilloscope define this spectral view.

B.3 High-frequency limit observation

For an additional high-frequency experiment, I used a much smaller hold capacitor and adjusted the conversion-clock divider. The signal generator was set to approximately 12.950 kHz, and the VGA display showed the resulting sparse, distorted waveform.

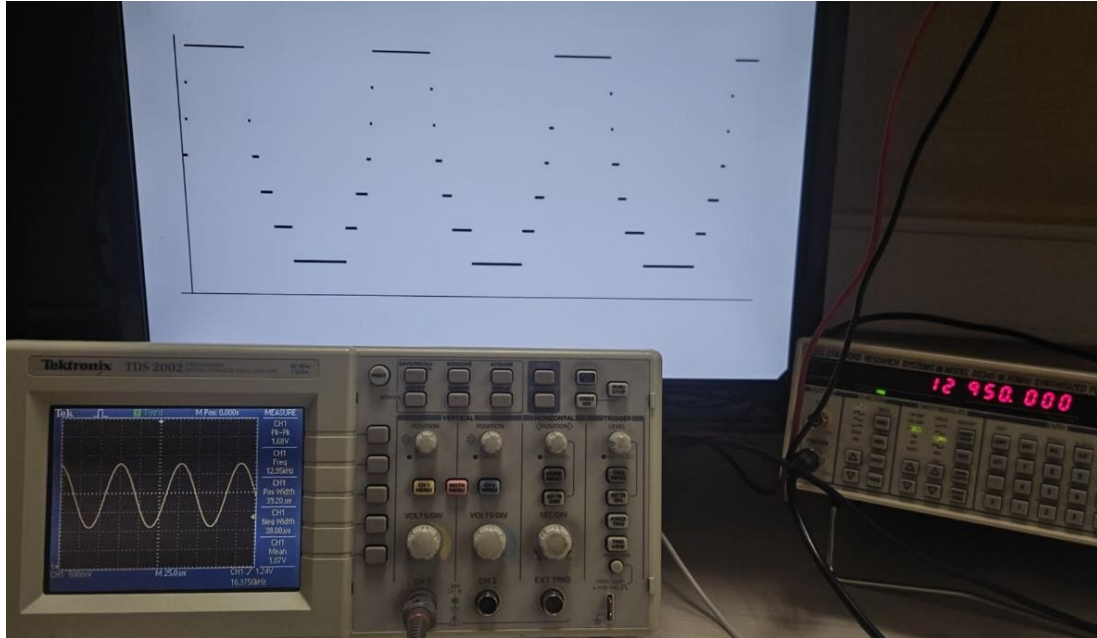


Figure 32. 12.950 kHz experiment with adjusted capacitor and clock-divider settings.

The sparse output shows how the displayed waveform changes as the analog acquisition time and conversion cadence approach the input period. This additional experiment explores the high-frequency behavior of the breadboard circuit.

Together, these laboratory results show the converter operating across the tested input frequencies and document the reference voltage, held-waveform spectrum, and high-frequency display behavior.

Appendix C Project Files and Laboratory Setup

C.1 Design files

The design files connect the circuit schematics to the FPGA implementation. The LTspice files contain the analog model and transient results, and the VHDL modules describe conversion, triggering, sample storage, and VGA rendering.

Table 20. Files associated with the circuit, FPGA, and laboratory presentation.

File or folder	Contents	Project role
top_main.vhd	Top-level module connections	Basys3 integration
top_adc.vhd	Conversion and trigger state machines	Breadboard ADC control

File or folder	Contents	Project role
vga_adc_scope_3bits.vhd	Capture buffers and VGA rendering	Waveform display
clk_wiz_0.xci	Pixel-clock configuration	25 MHz VGA clock
ecdproje.asc and ecdproje.raw	Analog schematic and transient results	Circuit simulation
2N7000.txt and BS250.txt	MOSFET models	LTspice device behavior
xdc.xdc	Clock and connector pin assignments	FPGA electrical interface
documentation/evidence/images	Circuit and laboratory figures	Schematics and demonstration results

C.2 Laboratory presentation setup

The analog circuit and reference network were assembled on solderless breadboards and connected to the Basys3 PMOD interface. The signal generator supplied the test input, the oscilloscope displayed the waveform and sample-and-hold behavior, and the VGA monitor showed the converted samples.

The nominal test used a 1 kHz waveform with 2 V peak-to-peak amplitude and a 3.5 V offset. The presentation included sine, square, triangle, and sawtooth inputs, followed by 500 Hz and 200 Hz sine waves. The oscilloscope, generator display, and VGA output appear together in the laboratory photographs in Chapter 11 and Appendix B.